

Mode 5 — Square-BFP (centered / opt)

DP32 (8×4) → DP16 (8×4) → DP8 (8×4) → block exponents → DP8 (4×4) → centered aligned add-and-square

Step 1 — DP32 (8×4)

$$\text{DP32}(8 \times 4) = \sum_{i=0}^{31} a_i b_i$$

Step 2 — DP16 (8×4)

$$\text{DP32}(8 \times 4) = \underbrace{\sum_{i=0}^{15} a_i b_i}_{\text{DP16}(8 \times 4)} + \underbrace{\sum_{i=16}^{31} a_i b_i}_{\text{DP16}(8 \times 4)}$$

Step 3 — DP8 (8×4)

$$\text{DP32}(8 \times 4) = \underbrace{\sum_{i=0}^7 a_i b_i}_{\text{DP8}(8 \times 4)} + \underbrace{\sum_{i=8}^{15} a_i b_i}_{\text{DP8}(8 \times 4)} + \underbrace{\sum_{i=16}^{23} a_i b_i}_{\text{DP8}(8 \times 4)} + \underbrace{\sum_{i=24}^{31} a_i b_i}_{\text{DP8}(8 \times 4)}$$

Step 4 — Block exponents (one exponent block per DP8 (8×4))

$$\left(\sum_{i=8q}^{8q+7} a_i b_i, E_q \right)_{q=0..3} \longrightarrow \left(M = \sum_{q=0}^3 2^{-\delta_q} \sum_{i=8q}^{8q+7} a_i b_i, E \right) \quad E_q = E_{a,q} + E_{b,\lfloor q/2 \rfloor}$$

$E = \max(E_0, \dots, E_3)$, $\delta_q = E - E_q$ (staged in the tree, the stage shifts telescope); truncating shifts, on block sums only;
 $\text{DP32}(8 \times 4) = M \cdot 2^E$.

Step 5 — DP8 (4×4)

$$\begin{aligned} M = & 2^{-\delta_0} \left(\underbrace{2^4 \sum_{i=0}^7 a_{H,i} b_i}_{\text{DP8}(4 \times 4)} + \underbrace{\sum_{i=0}^7 a_{L,i} b_i}_{\text{DP8}(4 \times 4)} \right) + 2^{-\delta_1} \left(\underbrace{2^4 \sum_{i=8}^{15} a_{H,i} b_i}_{\text{DP8}(4 \times 4)} + \underbrace{\sum_{i=8}^{15} a_{L,i} b_i}_{\text{DP8}(4 \times 4)} \right) \\ & + 2^{-\delta_2} \left(\underbrace{2^4 \sum_{i=16}^{23} a_{H,i} b_i}_{\text{DP8}(4 \times 4)} + \underbrace{\sum_{i=16}^{23} a_{L,i} b_i}_{\text{DP8}(4 \times 4)} \right) + 2^{-\delta_3} \left(\underbrace{2^4 \sum_{i=24}^{31} a_{H,i} b_i}_{\text{DP8}(4 \times 4)} + \underbrace{\sum_{i=24}^{31} a_{L,i} b_i}_{\text{DP8}(4 \times 4)} \right) \end{aligned}$$

Step 6 — Centered aligned add-and-square (subtract 8 from unsigned nibbles; per DP8 (4×4) add $c = 0/512/2048$ for 0/1/2 unsigned nibbles)

$$\begin{aligned} M = & 2^{-\delta_0} 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{H,i} + b_i)^2 - \sum_{i=0}^7 a_{H,i}^2 - \sum_{i=0}^7 b_i^2 \right)}_{\text{DP8}(4 \times 4)} \\ & + 2^{-\delta_0} \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 ((a_{L,i} - 8) + b_i)^2 - \sum_{i=0}^7 (a_{L,i} - 8)^2 - \sum_{i=0}^7 (b_i - 8)^2 + 512 \right)}_{\text{DP8}(4 \times 4)} \\ & + 2^{-\delta_1} 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} (a_{H,i} + b_i)^2 - \sum_{i=8}^{15} a_{H,i}^2 - \sum_{i=8}^{15} b_i^2 \right)}_{\text{DP8}(4 \times 4)} \\ & + 2^{-\delta_1} \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} ((a_{L,i} - 8) + b_i)^2 - \sum_{i=8}^{15} (a_{L,i} - 8)^2 - \sum_{i=8}^{15} (b_i - 8)^2 + 512 \right)}_{\text{DP8}(4 \times 4)} \\ & + 2^{-\delta_2} 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=16}^{23} (a_{H,i} + b_i)^2 - \sum_{i=16}^{23} a_{H,i}^2 - \sum_{i=16}^{23} b_i^2 \right)}_{\text{DP8}(4 \times 4)} \end{aligned}$$

$$\begin{aligned}
& + 2^{-\delta_2} \underbrace{\frac{1}{2} \left(\sum_{i=16}^{23} ((a_{L,i}-8)+b_i)^2 - \sum_{i=16}^{23} (a_{L,i}-8)^2 - \sum_{i=16}^{23} (b_i-8)^2 + 512 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{-\delta_3} \underbrace{2^4 \frac{1}{2} \left(\sum_{i=24}^{31} (a_{H,i}+b_i)^2 - \sum_{i=24}^{31} a_{H,i}^2 - \sum_{i=24}^{31} b_i^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^{-\delta_3} \underbrace{\frac{1}{2} \left(\sum_{i=24}^{31} ((a_{L,i}-8)+b_i)^2 - \sum_{i=24}^{31} (a_{L,i}-8)^2 - \sum_{i=24}^{31} (b_i-8)^2 + 512 \right)}_{\text{DP8 (4}\times\text{4)}}
\end{aligned}$$

Step 7 — All components

$$\begin{aligned}
\text{PE} &= 2^{-\delta_0} \left(2^4 \sum_{i=0}^7 (a_{H,i}+b_i)^2 + \sum_{i=0}^7 ((a_{L,i}-8)+b_i)^2 \right) + 2^{-\delta_1} \left(2^4 \sum_{i=8}^{15} (a_{H,i}+b_i)^2 + \sum_{i=8}^{15} ((a_{L,i}-8)+b_i)^2 \right) \\
&+ 2^{-\delta_2} \left(2^4 \sum_{i=16}^{23} (a_{H,i}+b_i)^2 + \sum_{i=16}^{23} ((a_{L,i}-8)+b_i)^2 \right) + 2^{-\delta_3} \left(2^4 \sum_{i=24}^{31} (a_{H,i}+b_i)^2 + \sum_{i=24}^{31} ((a_{L,i}-8)+b_i)^2 \right) \\
\alpha &= 2^{-\delta_0} \left(2^4 \sum_{i=0}^7 a_{H,i}^2 + \sum_{i=0}^7 (a_{L,i}-8)^2 \right) + 2^{-\delta_1} \left(2^4 \sum_{i=8}^{15} a_{H,i}^2 + \sum_{i=8}^{15} (a_{L,i}-8)^2 \right) \\
&+ 2^{-\delta_2} \left(2^4 \sum_{i=16}^{23} a_{H,i}^2 + \sum_{i=16}^{23} (a_{L,i}-8)^2 \right) + 2^{-\delta_3} \left(2^4 \sum_{i=24}^{31} a_{H,i}^2 + \sum_{i=24}^{31} (a_{L,i}-8)^2 \right) \\
\beta &= 2^{-\delta_0} \left(2^4 \sum_{i=0}^7 b_i^2 + \sum_{i=0}^7 (b_i-8)^2 \right) + 2^{-\delta_1} \left(2^4 \sum_{i=8}^{15} b_i^2 + \sum_{i=8}^{15} (b_i-8)^2 \right) \\
&+ 2^{-\delta_2} \left(2^4 \sum_{i=16}^{23} b_i^2 + \sum_{i=16}^{23} (b_i-8)^2 \right) + 2^{-\delta_3} \left(2^4 \sum_{i=24}^{31} b_i^2 + \sum_{i=24}^{31} (b_i-8)^2 \right) \\
C &= 2^{-\delta_0} (0 + 512) + 2^{-\delta_1} (0 + 512) + 2^{-\delta_2} (0 + 512) + 2^{-\delta_3} (0 + 512) \\
M &= \frac{1}{2} (\text{PE} - \alpha - \beta + C), \quad E = \max(E_0, E_1, E_2, E_3)
\end{aligned}$$